

- 1 Describe how to separate the following. In each example, give a description of the procedure used and explain why this method works.

(a) Copper powder from a mixture containing copper and zinc powders.

procedure

.....

explanation

.....

[3]

(b) Nitrogen from a mixture of nitrogen and oxygen.

procedure

.....

explanation

.....

[3]

(c) Glycine from a mixture of the two amino acids glycine and alanine. Glycine has the lower R_f value.

procedure

.....

explanation

.....

[2]

(d) Magnesium hydroxide from a mixture of magnesium hydroxide and zinc hydroxide.

procedure

.....

explanation

.....

[3]

[Total: 11]

2

An important aspect of chemistry is purity and methods of purification.

(a) Give an example of a substance used in everyday life which must be pure.

..... [1]

(b) A list of techniques used to separate mixtures is given below.

chromatography	crystallisation	diffusion
evaporation	filtration	fractional distillation
simple distillation		

(i) From the list, choose the most suitable technique to separate the following.

water from sea-water

helium from a mixture of helium and methane

ethanol from a mixture of ethanol and propanol

iron filings from a mixture of iron filings and water

a mixture of two amino acids, glycine and alanine

[5]

(ii) Describe how you would obtain a pure sample of copper(II) sulfate-5-water crystals from a mixture of copper(II) sulfate-5-water with copper(II) oxide using some of the techniques listed above.

.....

.....

.....

.....

..... [4]

[Total: 10]

3 Petroleum contains hydrocarbons which are separated by fractional distillation.

(a) (i) Complete the following definition of a hydrocarbon.

A hydrocarbon is a compound which
..... [2]

(ii) Explain what is meant by the term *fractional distillation*.

.....
.....
..... [2]

(b) Some of the fractions obtained from petroleum are given below.
State a use for each fraction.

bitumen

lubricating fraction

paraffin fraction

gasoline fraction [4]

[Total: 8]

4 A list of techniques used to separate mixtures is given

below.

- filtration
- diffusion
- fractional distillation
- simple distillation
- crystallisation
- chromatography

From this list, choose the most suitable technique to separate the following mixtures.
A technique may be used once, more than once or not at all.

- (a)** butane from a mixture of propane and butane [1]
- (b)** oxygen from liquid air [1]
- (c)** water from aqueous magnesium sulfate [1]
- (d)** potassium chloride from aqueous potassium chloride [1]
- (e)** silver chloride from a mixture of silver chloride and water [1]
- (f)** glucose from a mixture of glucose and maltose [1]

[Total: 6]

5 The following techniques are used to separate mixtures.

A simple distillation

B fractional distillation

C evaporation

D chromatography

E filtration

F diffusion

From this list, choose the most suitable technique to separate the following.

(a) methane from a mixture of the gases, methane and ethane [1]

(b) water from aqueous magnesium sulfate [1]

(c) glycine from a mixture of the amino acids, glycine and lysine [1]

(d) iron filings from a mixture of iron filings and water [1]

(e) zinc sulfate crystals from aqueous zinc sulfate [1]

(f) hexane from a mixture of the liquids, hexane and octane [1]

[Total: 6]

- 4** A sample of muddy river water contains water, dissolved solids and insoluble solid mud.

Plan an investigation to find the concentration of dissolved solids, in g/dm^3 , in the river water.

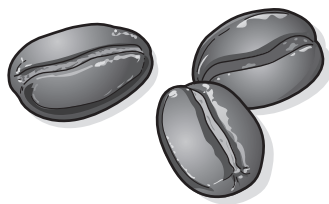
In your answer state how you will work out the concentration of the dissolved solids in g/dm^3 .

You are provided with a small sample (less than 1 dm^3) of muddy river water and common laboratory apparatus.

(1 dm³ = 1000 cm³)

[6]

4 The diagram shows some coffee beans.



Caffeine occurs naturally in coffee beans. Caffeine is a white crystalline solid. It is very soluble in hot water but much less soluble in cold water.

Plan an investigation to obtain a pure crystalline sample of caffeine from coffee beans.

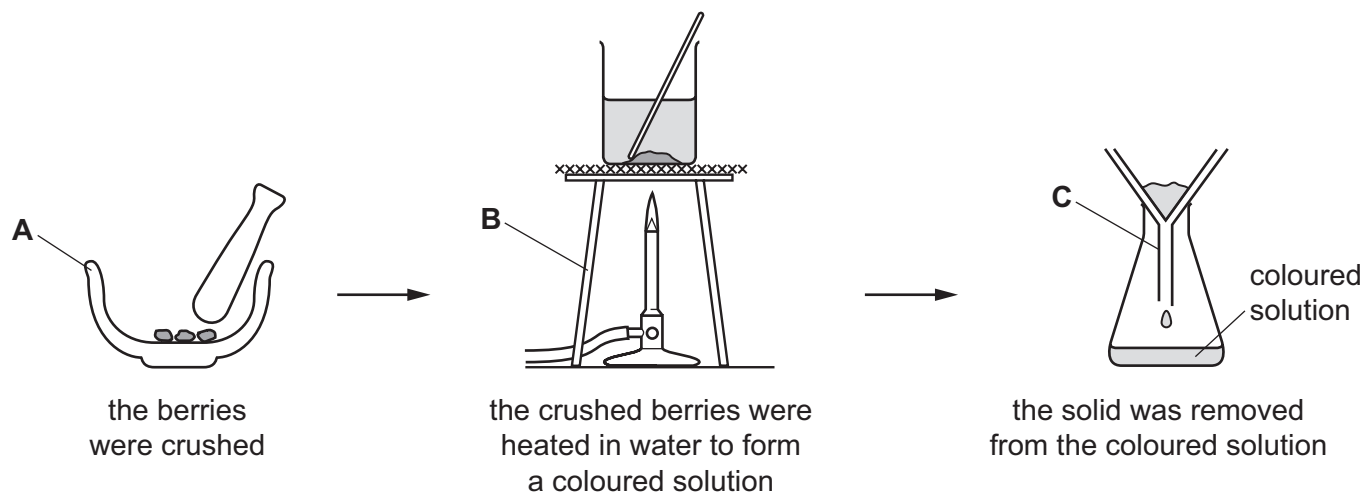
Assume that all other soluble substances in coffee beans are very soluble in both hot and cold water.

You are provided with coffee beans and common laboratory apparatus.

[6]

- 1 Many indicators are coloured substances obtained from plants.

A student extracted the coloured substances from some berries using the method shown.



- (a) Name the items of apparatus labelled **A**, **B** and **C**.

A

B

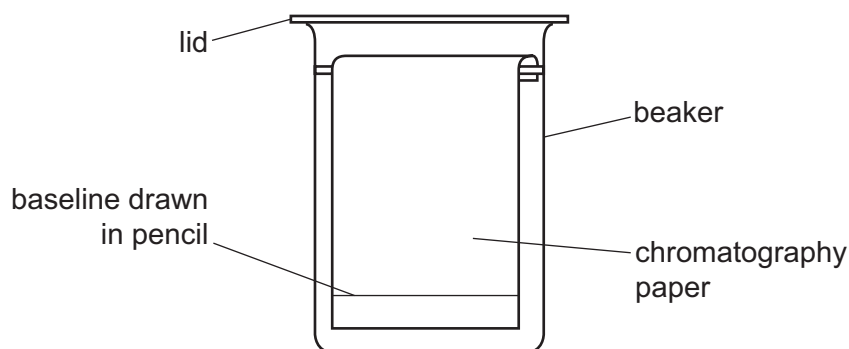
C

[3]

- (b) The student analysed the coloured solution using chromatography.

- (i) Complete the diagram to show:

- where the spot of coloured solution should be placed on the paper
- the level of the solvent in the beaker.



[2]

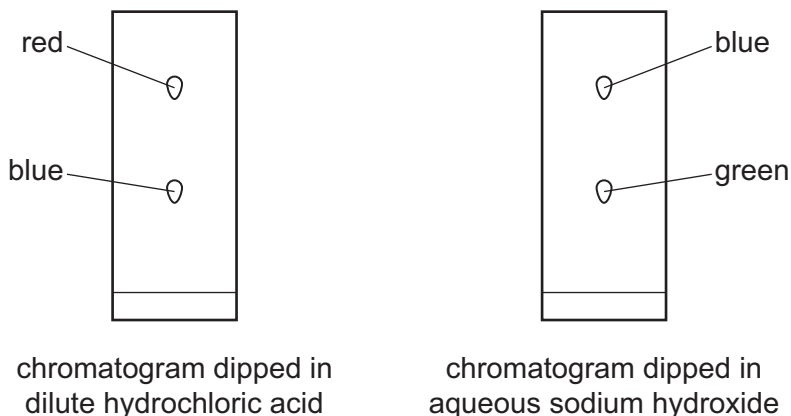
- (ii) Explain why pencil is used to draw the baseline on the chromatography paper.

.....

..... [1]

- (c) The student made two chromatograms. After chromatography, one chromatogram was dipped in dilute hydrochloric acid and one was dipped in aqueous sodium hydroxide.

The results are shown.



- (i) Determine the number of coloured substances in the solution obtained from the berries.

..... [1]

- (ii) The table gives the colours of some indicators in acid and alkali.

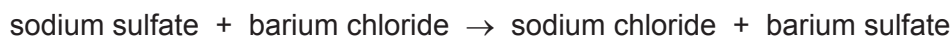
name of indicator	colour in acid	colour in alkali
anthocyanin	red	blue
bromothymol blue	yellow	blue
congo red	blue	red
methyl purple	purple	green

Use the data in the table and the results to give a possible identity for **one** indicator in the berries.

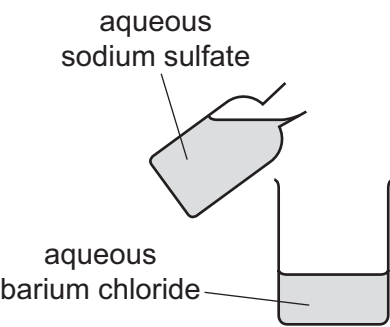
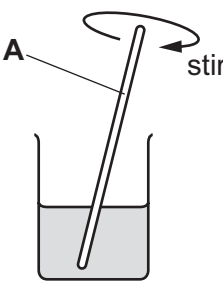
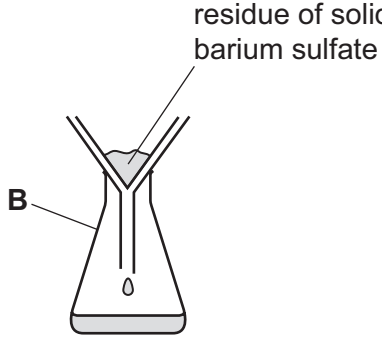
..... [1]

[Total: 8]

- 1 Barium sulfate is an insoluble salt. Barium sulfate can be made by reacting excess aqueous sodium sulfate with aqueous barium chloride.



A student made a sample of barium sulfate using the following steps.

step 1	step 2	step 3
 aqueous sodium sulfate aqueous barium chloride excess aqueous sodium sulfate was added to aqueous barium chloride	 A stir the mixture was stirred	 residue of solid barium sulfate B the solid barium sulfate was removed from the mixture

- (a) Name the items of apparatus labelled **A** and **B**.

A

B

[2]

- (b) Name the process shown in **step 3**.

..... [1]

- (c) The general name for the solid in **step 3** is residue.

State the general name for the solution obtained from the process in **step 3**.

..... [1]

- (d) Two more steps, **step 4** and **step 5**, are needed to obtain a pure sample of barium sulfate. In each of these steps something is removed from the residue.

State what is done in each of **step 4** and **step 5** and identify the substance removed from the barium sulfate.

step 4

.....

substance removed

step 5

.....

substance removed

[4]

[Total: 8]

4 Tartrazine is used as a yellow food colouring.

Plan an investigation to find out if a yellow sweet contains tartrazine. Explain how your results will tell you if the sweet contains tartrazine.

You have access to all normal laboratory materials, a yellow sweet and a sample of tartrazine.

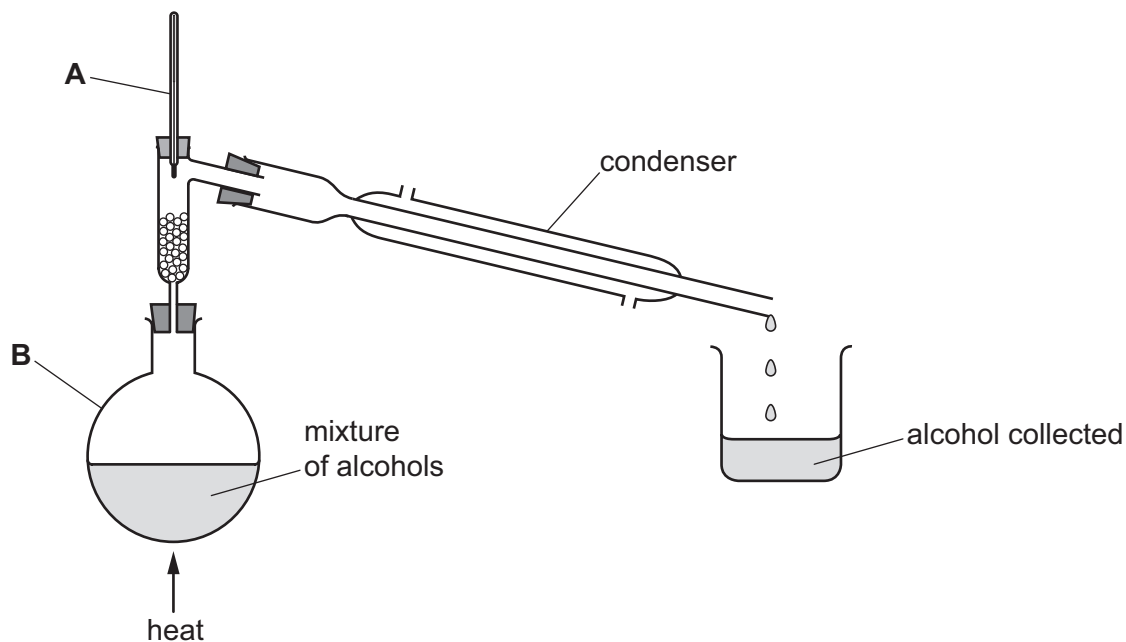
You may draw a labelled diagram as part of your answer.

[6]

1 The table gives the boiling points of four alcohols.

alcohol	boiling point/°C
methanol	65
ethanol	79
propan-1-ol	97
butan-1-ol	117

The apparatus shown can be used to separate a mixture of the four alcohols shown in the table.



(a) Name the apparatus labelled **A** and **B**.

A

B

[2]

(b) Add to the diagram **one** arrow to show where water enters the condenser.

[1]

(c) (i) Why is it **not** safe to heat the mixture of alcohols with a Bunsen burner?

..... [1]

(ii) Suggest how the mixture of alcohols can be heated safely?

..... [1]

(d) Describe how the condenser allows the alcohol to be collected as a liquid.

.....
..... [1]

(e) Which alcohol would be collected first?
Explain your answer.

alcohol collected first

explanation

..... [2]

[Total: 8]

- 4** A black dye can be obtained from some plant roots.

Plan an investigation to determine how many different coloured substances are contained in a black dye obtained from plant roots.

You must include how the results you obtain will tell you how many different coloured substances are contained in the black dye.

You have access to plant roots and all normal laboratory apparatus.

[6]

- 4 (a) Filtration and chlorination are two stages in water treatment.

State the purpose of each stage.

filtration

.....

chlorination

.....

[2]

- (b) A student uses anhydrous copper(II) sulfate to test for the presence of water.

- (i) What colour change is seen if water is present?

from to [2]

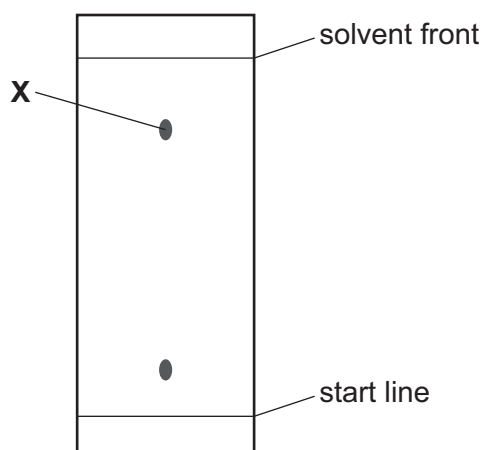
- (ii) The purity of a sample of water can be assessed by measuring its boiling point.

How is the boiling point of water affected by impurities?

..... [1]

- (c) Chromatography can be used to test the purity of substances.

The diagram shows the chromatogram of a coloured substance.



- (i) How does this chromatogram show that this substance is **not** pure?

..... [1]

- (ii) Draw a circle round the correct R_f value for the spot labelled **X**.

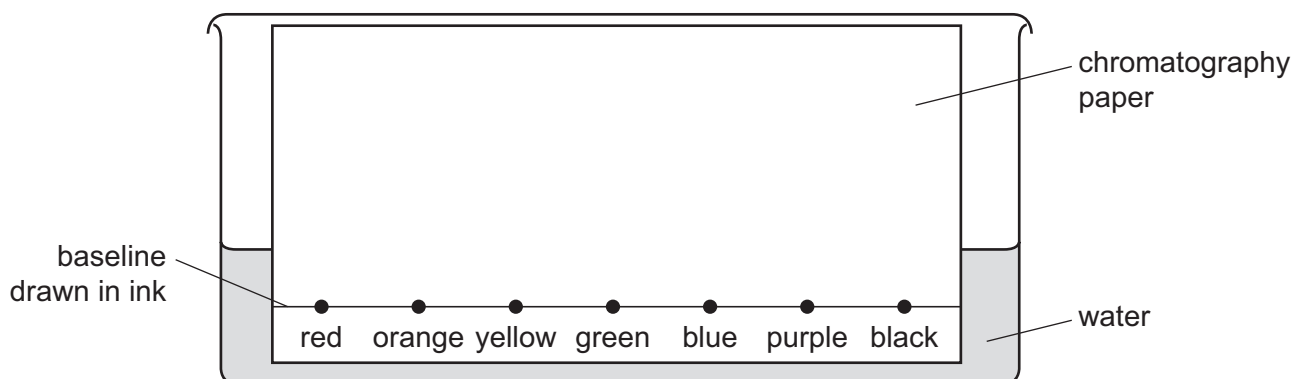
0.2 0.4 0.8 1.2 [1]

- (iii) State how a colourless substance can be made visible on a chromatogram.

..... [1]

[Total: 8]

- 1 A student investigated the dyes contained in different coloured inks using chromatography. Water was the solvent. The diagram shows how the student set up the apparatus.



- (a) Identify **two** errors in the way the student set up the apparatus.

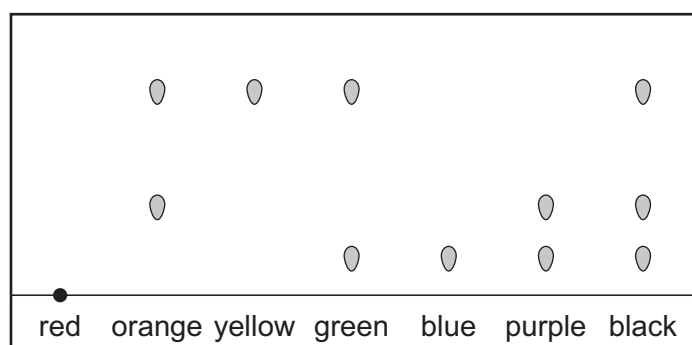
1

2

[2]

- (b) The student then carried out the chromatography correctly.

The diagram shows the results.



- (i) Which ink contains the greatest number of soluble dyes?

..... [1]

- (ii) Which **two** inks are made of a single soluble dye?

..... and

[1]

- (iii) From the chromatogram it is **not** possible to tell if the red ink contains different dyes.

Suggest how the experiment could be changed to find out if the red ink contains different dyes.

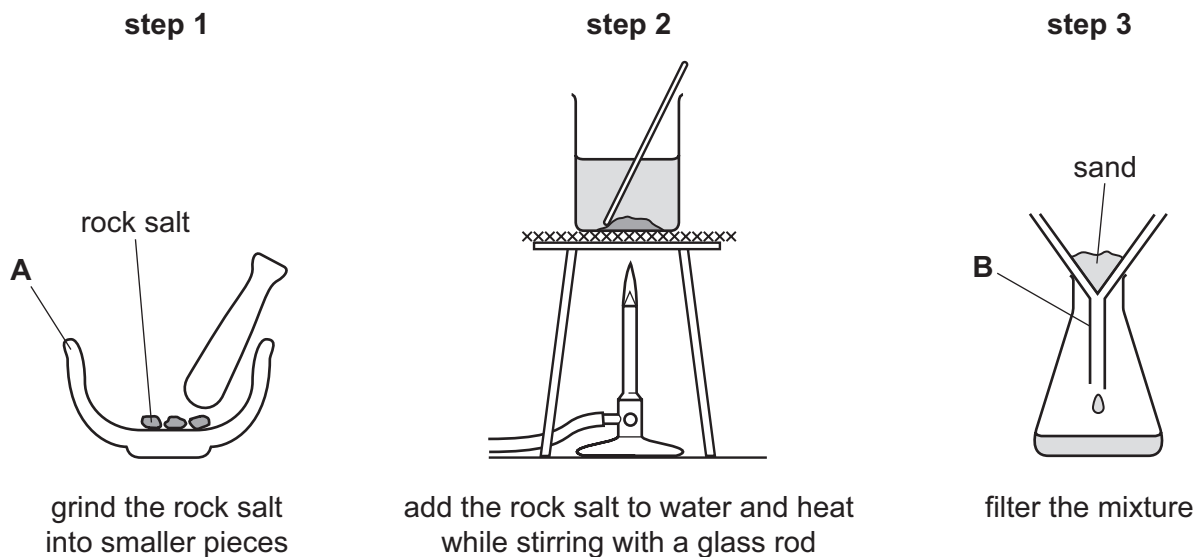
..... [1]

[Total: 5]

- 1 A sample of rock salt contains sodium chloride and sand.

Sodium chloride is soluble in water. Sand is insoluble in water.

A student obtained dry crystals of pure sodium chloride from a lump of rock salt. These are some of the steps the student used.



- (a) Name the apparatus labelled **A** in **step 1**.

..... [1]

- (b) Explain why the mixture is heated and stirred in **step 2**.

..... [1]

- (c) (i) Name the apparatus labelled **B** in **step 3**.

..... [1]

- (ii) State the scientific term for the sand left on the filter paper in **step 3**.

..... [1]

- (d) Describe what the student must do after **step 3** to obtain dry crystals of pure sodium chloride.

.....

.....

.....

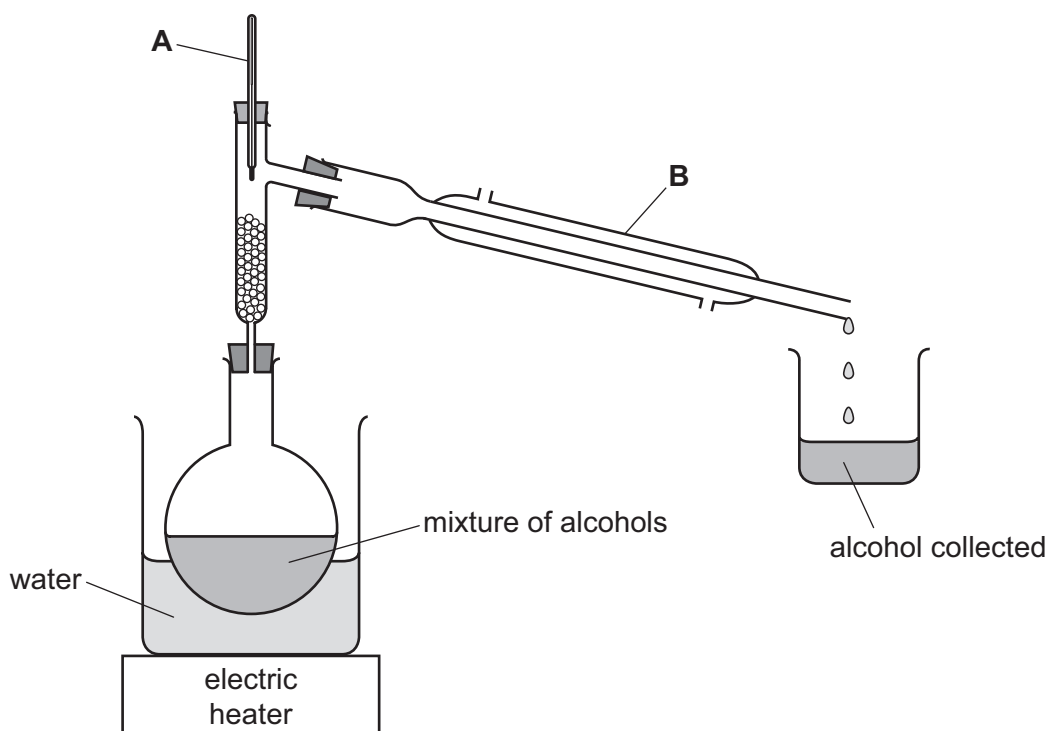
..... [3]

[Total: 7]

- 1 The table gives the boiling points of four alcohols.

alcohol	boiling point/°C
butanol	117
ethanol	79
pentanol	138
propanol	97

The apparatus shown can be used to obtain propanol from a mixture containing butanol, ethanol, pentanol and propanol.



- (a) Name the items of apparatus labelled **A** and **B**.

A

B

[2]

- (b) Name this method of separation.

..... [2]

- (c) Explain why it is safer to heat the mixture of alcohols in the way shown rather than with a Bunsen burner.

..... [1]

(d) Describe how propanol can be obtained from the mixture. Use data from the table.

.....

.....

..... [2]

(e) Explain why the apparatus in the diagram **cannot** be used to obtain butanol from the mixture.

.....

..... [1]

[Total: 8]

- 4 A mixture contains three solid compounds:
- copper(II) sulfate
 - cetyl alcohol
 - silicon dioxide.

The table gives some information on the solubility of these three solids.

name of compound	solubility in water	solubility in propanone
copper(II) sulfate	soluble	insoluble
cetyl alcohol	insoluble	soluble
silicon dioxide	insoluble	insoluble

Plan a method to obtain a pure sample of each of the three solids, copper(II) sulfate, cetyl alcohol and silicon dioxide, from the mixture.

You have access to normal laboratory apparatus.

[6]

- 4 A toothpaste contains:
- sodium fluoride
 - calcium carbonate
 - silica
 - mint flavouring.

Sodium fluoride and the mint flavouring are soluble in water.

Calcium carbonate and silica are insoluble in water.

Calcium carbonate reacts with dilute hydrochloric acid to form the soluble salt calcium chloride.

Plan an investigation to find the percentage by mass of silica in the toothpaste.

In your answer you should include how you will calculate the percentage by mass of silica in the toothpaste.

You have access to normal laboratory apparatus.

[6]

1 A student did the following steps to make zinc chloride crystals from solid zinc oxide.

- step 1** Pour 40 cm³ of dilute hydrochloric acid into a beaker. Add a small amount of zinc oxide. Warm the mixture and stir it.
- step 2** Continue to add zinc oxide to the beaker until all of the dilute hydrochloric acid has reacted.
- step 3** Remove the excess zinc oxide.
- step 4** Obtain crystals of zinc chloride from the solution.

(a) Name the apparatus used in **step 1** to:

(i) add the zinc oxide

..... [1]

(ii) warm the mixture.

..... [1]

(b) How did the student know that all of the dilute hydrochloric acid had reacted in **step 2**?

.....
 [1]

(c) (i) What is meant by the term *excess* in **step 3**?

..... [1]

(ii) How is the excess zinc oxide removed in **step 3**?

..... [1]

(d) Describe how the crystals are obtained in **step 4**.

.....

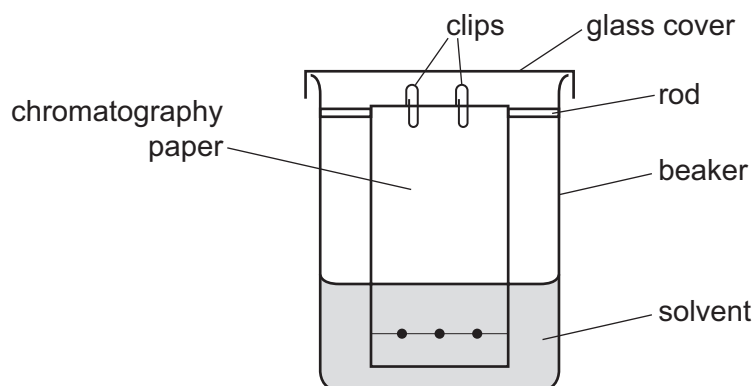
 [3]

(e) Suggest how the method would differ if zinc carbonate were used instead of zinc oxide.

..... [1]

[Total: 9]

- 1 A student investigated the colours present in three hair dyes, **P**, **Q** and **R**, using chromatography. **P**, **Q** and **R** are insoluble in water. The student suggested setting up the apparatus for the experiment as shown.



- (a) Why is a lid necessary on top of the beaker?

..... [1]

- (b) (i) Identify **one** mistake in the student's diagram.

..... [1]

- (ii) Suggest why this mistake would stop the experiment working.

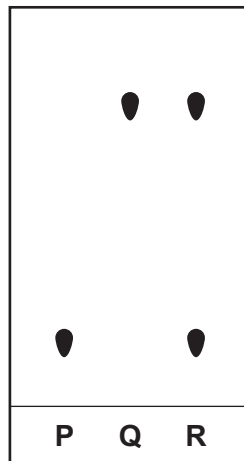
.....

..... [1]

- (c) Name a suitable solvent that could be used in this experiment.

..... [1]

- (d) A separate chromatography experiment was done using the hair dyes **P**, **Q** and **R**. The chromatogram obtained is shown.



State **three** conclusions about the hair dyes **P**, **Q** and **R** which can be deduced from the chromatogram.

1

2

3

[3]

[Total: 7]

4 Iodine dissolves in two different solvents: ethanol and hexane.

Plan an experiment to find out in which solvent iodine is the most soluble at room temperature.

You are provided with iodine, the two solvents and common laboratory apparatus.

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..... [6]

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- 4 The table gives some information about the properties of three substances found in a hand cream.

substance	reaction with dilute nitric acid
polystyrene beads	no reaction
calcium carbonate	reacts and dissolves
sodium fluoride	dissolves

Use the information in the table to plan an experiment to obtain a pure, dry sample of polystyrene beads from this mixture of substances.

You are provided with a mixture of the three substances and common laboratory apparatus.

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.....

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.....

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.....

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..... [6]